

3. Magnesium oxide has various applications in water treatment. It is an effective base, offering a safe method of changing pH compared to other neutralising agents such as caustic soda. It also helps prevent scale formation in boilers, heat exchangers, and piping.

(a) State **one** adverse effect of scale on household plumbing. [1]

(b) Burning magnesium in a gas jar of oxygen produces magnesium oxide on a small scale.



(i) Describe the atomic structure of magnesium. [3]

(ii) Draw a diagram to show the electronic structure of a magnesium atom. [1]

(iii) Write a balanced symbol equation for the reaction between magnesium and oxygen. [3]

(iv) State how the electron configuration of a magnesium atom changes when magnesium reacts with oxygen. [1]